Paper Modifications for the GAN HPO Experiment

All text changes required to add the GAN experiment to the BIF paper

Auto-generated from v3 experiment results

March 28, 2026

This document lists every paragraph that must be modified or added, in order. Yellow boxes indicate the exact location in the existing paper. New or replacement text is shown verbatim below each box.

Figure 4 composite: `figures/figure4_composite.pdf` (pre-built). Panel sources:

- Panel A: `panelA.pdf` — DCGAN architecture (standard task)
- Panel B: `v3_fair_main_ro.png` — Main RO
- Panel C: `v3_fair_main_r2.png` — Main Parent + Child R^2
- Panel D: `panelB.pdf` — DCGAN architecture (SpectralNorm task)
- Panel E: `v3_fair_modularity_ro.png` — Modularity RO
- Panel F: `v3_fair_modularity_r2.png` — Modularity Parent + Child R^2

Modification 1: Abstract

Abstract — last sentence before "1. Introduction".
Replace "...on synthetic and real-world neurostimulation optimization tasks."

...on synthetic, real-world neurostimulation, and realistic GAN hyperparameter optimization tasks.

Modification 2: Section 4 "Datasets" — opening sentence

Section 4 Datasets, first sentence.
Replace "three benchmarks" with "four benchmarks".

We evaluate BIF on four benchmarks designed to test hierarchical transfer and scalability. Both synthetic tasks include 10% additive Gaussian Noise.

Modification 3: Section 4 "Datasets" — add 4th bullet

Section 4 Datasets, after the Neurostimulation bullet point.
Add the following as a new bullet.

- *GAN Hyperparameter Optimization:* A tabular surrogate built from 2,401 DCGAN training runs on Fashion-MNIST, spanning a 7^4 grid over four hyperparameters split into two functional groups: generator (learning rate, weight decay) and discriminator (learning rate, dropout). Each configuration was trained to convergence and scored by Fréchet Inception Distance (FID). A second surrogate replaces the standard discriminator with a Spectral-Norm variant, sharing the generator architecture. This pair of surrogates enables testing BIF's modularity: the generator child trained on the standard task can be transferred to the Spectral-Norm task without retraining, since only the discriminator sub-problem changes.

Modification 4: Section 4 "Implementation Details" — append

Section 4 Implementation Details, end of paragraph (after "Results report the mean...").
Replace the last sentence and append GAN-specific details.

Results report the mean and standard error across 10 randomly initialized trials (30 for the GAN surrogate). For the GAN surrogate, we use $\kappa = 20$, $\gamma = 4$, $\nu = 1.5$, with a log-normal lengthscale prior centered at 1.0. All methods share the same 20-query parent budget; BIF and Laferrière each use 3 initialization queries counted in the budget.

Modification 5: Section 5 — new subsection

Section 5 Results, after "Real-World Application: Neurostimulation" and before "Modularity and Subtask Transfer".
Insert the following new subsection.

GAN Hyperparameter Optimization. We evaluate BIF on a tabular DCGAN surrogate (Section 4), where the generator and discriminator form natural child sub-problems within a joint 4D parent space. As summarized in Table 1 and Figure 4(a–c), BIF matches vanilla GPBO in Relative Optimum (96.1 ± 0.4 vs. 96.0 ± 0.4) while achieving the highest parent R^2 (11.4%) and child R^2 (27.7%) among all methods. Laferrière lags in both RO (94.7 ± 0.6) and child R^2 (23.1%), confirming that frozen children limit the hierarchy's ability to refine sub-task representations. Although parent R^2 remains modest across all methods due to the difficulty of modelling 2,401 points from only 20 queries, BIF's child R^2 rises steadily (Fig. 4c), driven by the response contribution system (Section 3). The Laferrière children, by contrast, plateau immediately after initialization, as no downward signal reaches them.

Modification 6: Section 5 "Modularity and Subtask Transfer" — append

Section 5 "Modularity and Subtask Transfer", after the existing two paragraphs.
Append the following new paragraph.

We further validate modularity on the GAN surrogate, where the generator architecture is shared between the standard and Spectral-Norm discriminator tasks. As shown in Figure 4(d–f) and Table 1, BIF hot-start transfers the pre-trained generator child and continues to refine both children on the new task, reaching 95.3 ± 0.5 RO and 45.6% child R^2. Laferrière hot-start carries the same generator child but freezes it, achieving only 93.3 ± 0.5 RO and 23.2% child R^2 — nearly half of BIF's child accuracy. Vanilla GPBO, starting from scratch without any transferred knowledge, reaches 93.5 ± 0.7 RO.

Modification 7: Section 6 "Conclusion" — update sentence

Section 6 Conclusion, second paragraph, first sentence.
Replace "Empirical evaluations on 2D and 3D synthetic benchmarks, as well as a real-world neurostimulation task..."

Empirical evaluations on synthetic benchmarks, a real-world neurostimulation task, and a realistic GAN hyperparameter optimization surrogate demonstrate that BIF significantly improves sample efficiency, with higher AUC across the board, and overall optimization performance.

Modification 8: Table 1 — add GAN rows

Table 1. Add the following two row groups at the bottom, after Neural Stimulation.
Also update the caption to note the GAN query/seed counts.

Dataset	Model	Relative Optimum	Parent R^2	Child R^2	AUC (RO)
	... existing rows (Synthetic 2D, 3D, Neural Stimulation) ...				
GAN HPO	Vanilla GPBO	96.0 ± 0.4	9.7 ± 1.0	–	95.1 ± 0.4
	Laferrière	94.7 ± 0.6	6.8 ± 1.1	23.1 ± 2.4	92.0 ± 0.4
	BIF	96.1 ± 0.4	11.4 ± 1.3	27.7 ± 2.9	94.9 ± 0.4
GAN HPO (Modularity)	Vanilla GPBO	93.5 ± 0.7	43.7 ± 1.0	–	91.9 ± 0.8
	Laferrière (hot)	93.3 ± 0.5	48.8 ± 1.0	23.2 ± 2.4	88.5 ± 0.7
	BIF (hot-start)	95.3 ± 0.5	48.5 ± 1.2	45.6 ± 2.6	91.0 ± 0.5

Table 1: *

Updated Table 1 caption: Summary of performance metrics. GAN results use 20 queries and 30 seeds; other datasets use 100 queries and 10 seeds. Vanilla GPBO has no child R^2 (non-hierarchical).

Modification 9: Figure 4 caption

New Figure 4. Pre-built composite: `figures/figure4_composite.pdf`.
Use the following caption.

Figure 4. GAN hyperparameter optimization experiment. **(a)** DCGAN architecture for the standard task (generator: learning rate × weight decay; discriminator: learning rate × dropout). **(b)** Relative Optimum on the standard task (blue: Vanilla; orange: Laferrière; green: BIF). **(c)** Parent and child R^2 on the standard task. **(d)** DCGAN architecture for the Spectral-Norm task (shared generator, modified discriminator). **(e)** Relative Optimum on the modularity task (Vanilla from scratch; Laferrière and BIF hot-start with transferred generator child). **(f)** Parent and child R^2 on the modularity task. Shaded regions denote ±1 SEM over 30 seeds.

Summary of all modifications

1. **Abstract**: “synthetic and real-world neurostimulation” → add “and GAN hyperparameter optimization”
2. **Section 4 Datasets**: “three benchmarks” → “four benchmarks”; add GAN bullet
3. **Section 4 Implementation Details**: append GAN hyperparameters; update trial count
4. **Section 5 Results**: insert “GAN Hyperparameter Optimization” subsection (after Neurostimulation)
5. **Section 5 Modularity**: append GAN modularity paragraph
6. **Section 6 Conclusion**: update empirical summary sentence
7. **Table 1**: add 6 rows (GAN HPO + GAN HPO Modularity); update caption
8. **Figure 4**: new 6-panel composite figure with caption